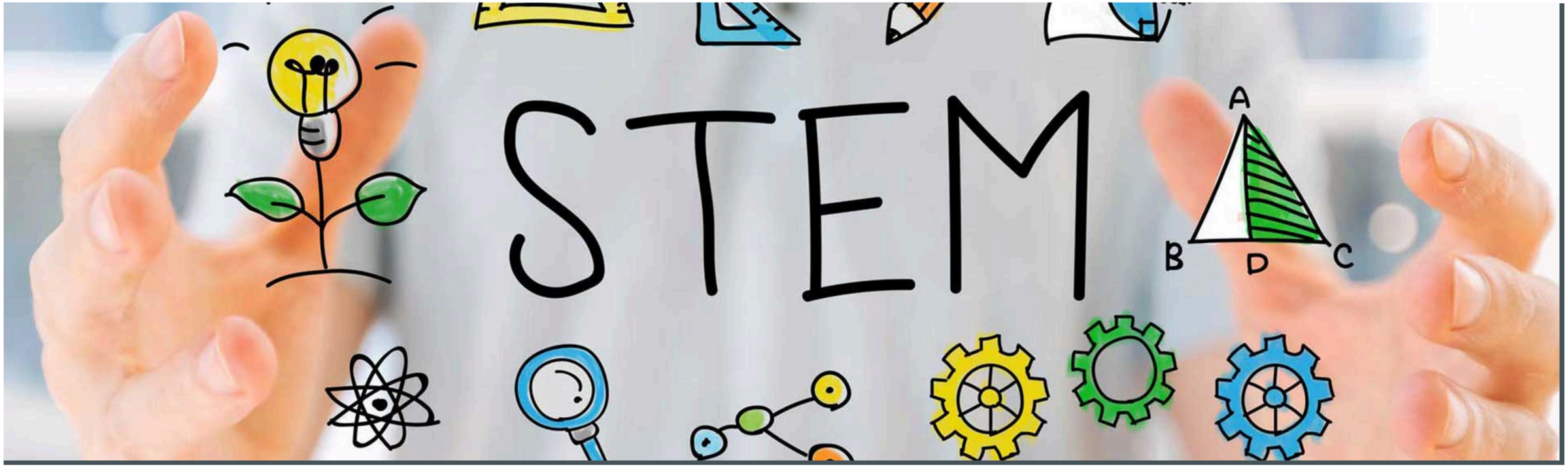
Study Buddy

STEM AI Integrated AI-Powered STEM Learning Ecosystem for Personalized, Adaptive & Multimodal Education

Project ID : R26-IT-009



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AI Integrated STEM Learning Ecosystem for Personalized, Adaptive & Multimodal Education



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The Problem

Traditional STEM education in Sri Lankan schools mainly relies on:

- Textbook-based learning
- One-way classroom teaching
- Memorization-focused assessments
- Limited personalized support

Impact of the Problem

- Low engagement in STEM subjects
- Difficulty understanding abstract scientific concepts
- Lack of adaptive learning support
- Stress during assessments
- Poor knowledge retention
- Limited access to personalized tutoring and revision assistance

Users & Stakeholders

- Students
- Teachers
- Parents / Guardians
- Schools & Educational Institutions
- Education Authorities

Reserch Gap & Objectives

Research Gap

- Existing educational systems have limitations in combining:
- Personalized STEM learning
- Context-aware conversational tutoring
- Multilingual and multimodal interaction
- Emotion-aware adaptive assessment
- Personalized knowledge synthesis
- Syllabus-grounded AI responses

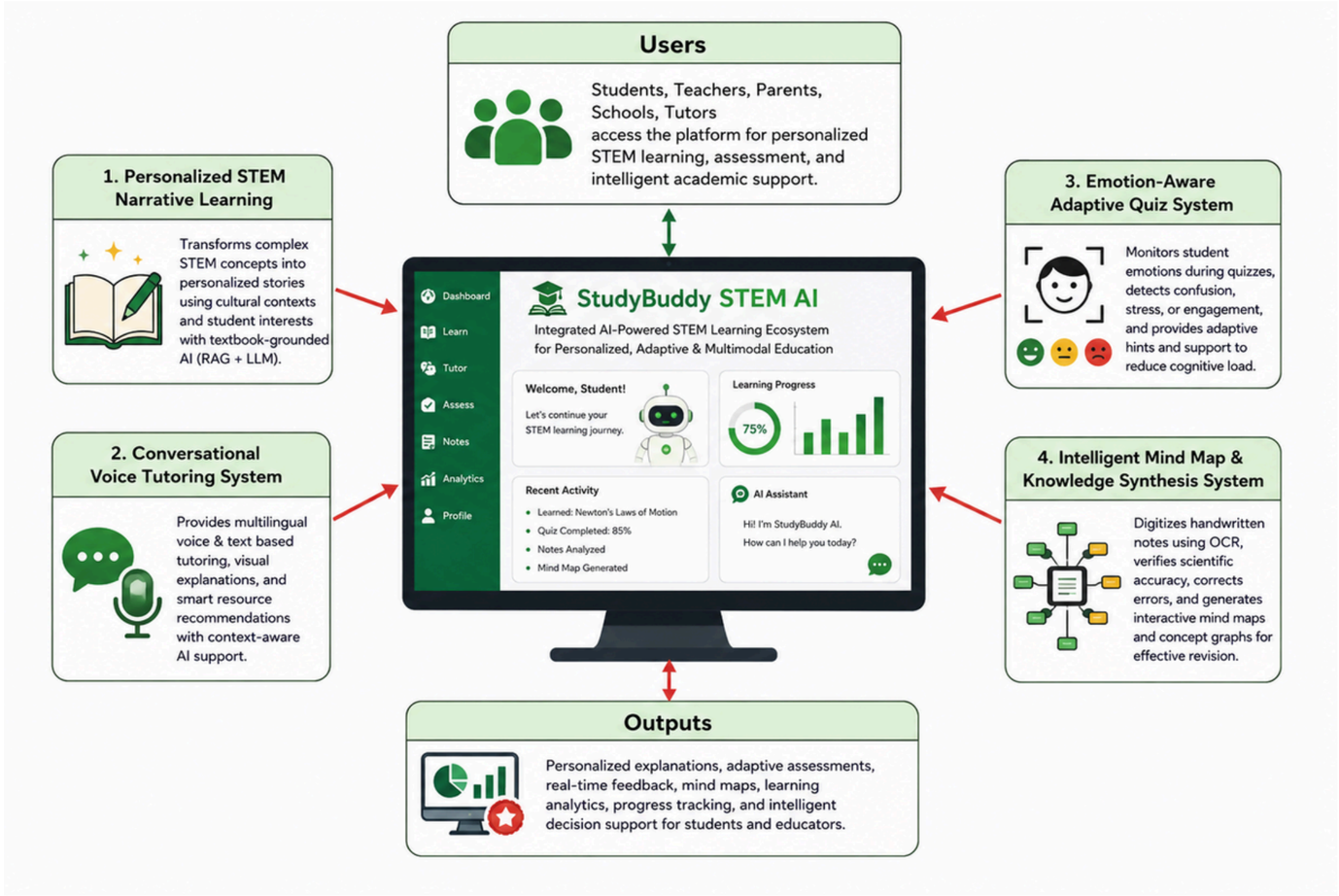
Key Objectives

- Provide personalized and adaptive STEM learning experiences.
- Support intelligent conversational STEM tutoring.
- Generate personalized STEM learning narratives.
- Adapt assessments based on student behaviour and learning state.
- Generate structured study materials from student content.
- Improve STEM engagement, understanding and knowledge retention.

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Solution Overview



1. RAG-Based Personalized STEM Storytelling

Generates interactive and personalized STEM stories using RAG to ensure accuracy, syllabus alignment, and real-world relevance.

Helps students understand concepts through engaging narratives and real-life scenarios.

2. Multilingual Conversational AI for STEM Tutoring

Provides context-aware tutoring through text, voice, and image inputs in multiple languages.

Supports concept explanation, doubt solving, and step-by-step learning guidance.

3. Emotion-Aware Adaptive Assessment

Detects student emotions and learning behavior to adapt question difficulty, provide personalized feedback, and recommend suitable learning resources.

Improves engagement and reduces assessment stress.

4. Handwritten Content Analysis & Knowledge Synthesis

Extracts handwritten notes using OCR and generates mind maps, flashcards, summaries, and concept verification.

Helps students revise better and convert notes into structured, easy-to-learn knowledge.

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Integration And Completion Status

COMPLETED / IMPLEMENTED

- Initial RAG pipelines
- Educational dataset preprocessing
- Conversational AI prototype
- Personalized storytelling prototype
- Emotion detection integration
- Adaptive assessment workflow
- Handwritten OCR processing
- Knowledge retrieval and verification
- Database and API development
- Initial frontend integration



BUILDING AN AI-POWERED PERSONALIZED STEM LEARNING ECOSYSTEM

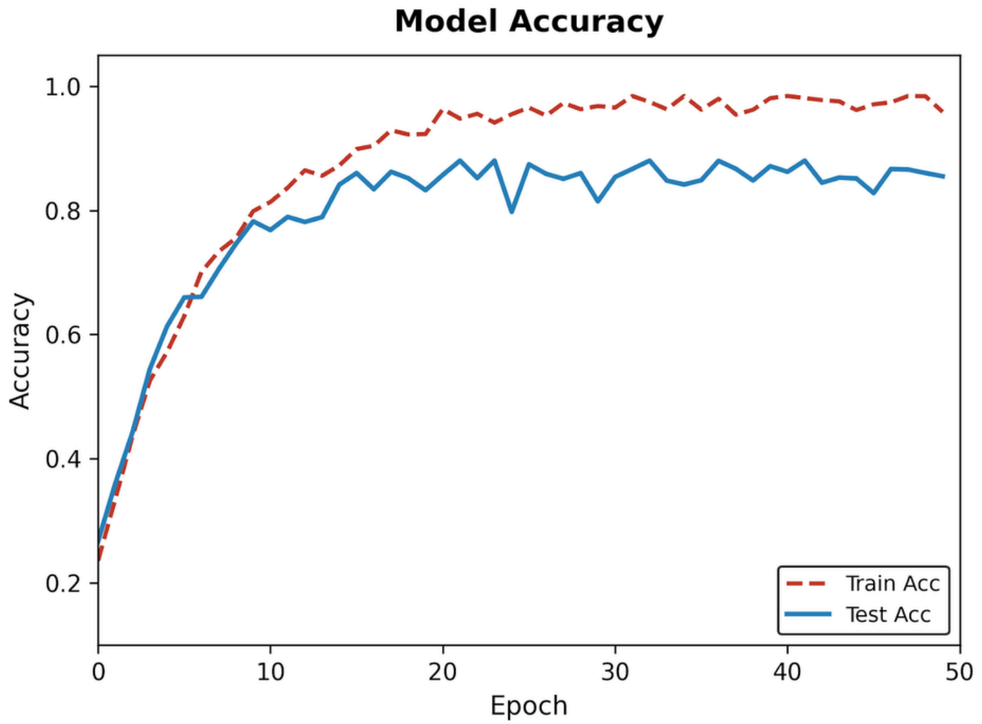
REMAINING WORK

- Final end-to-end integration
- Performance optimization
- Model refinement
- Integration validation
- UI/UX improvements
- Final system testing
- Deployment preparation

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Evaluation Evidence



Persona Classification Model

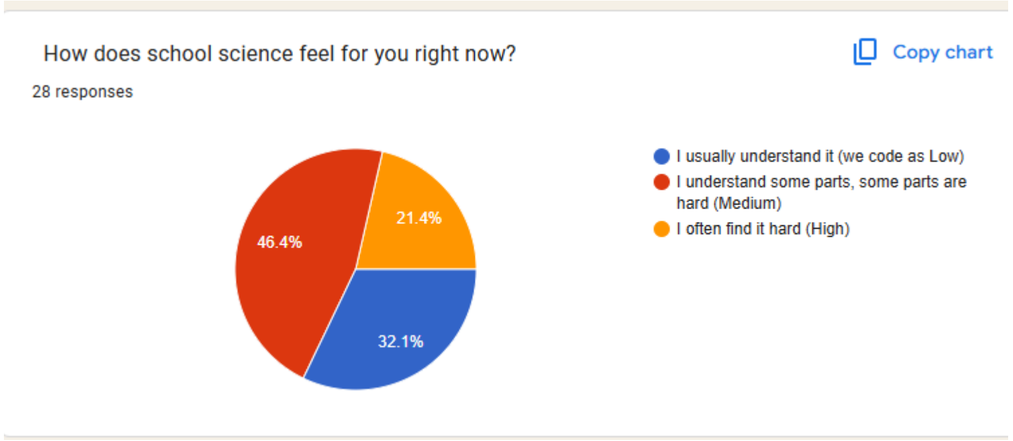
Evaluation Evidence

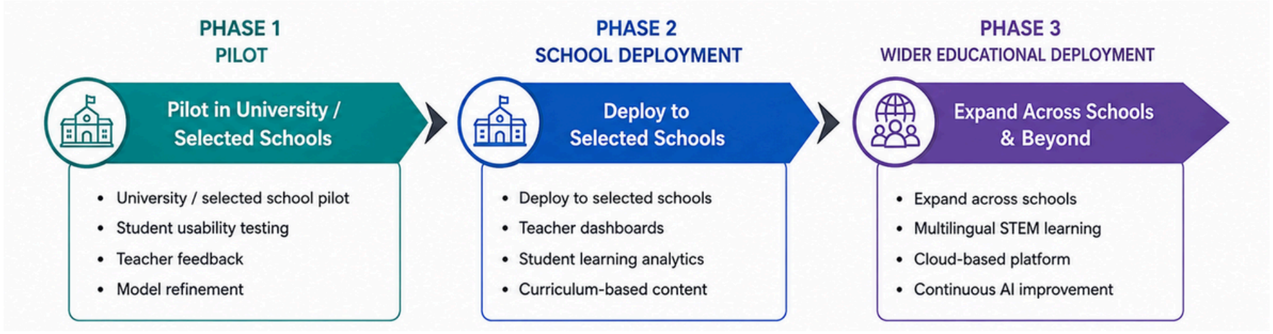
NFR	Evidence / How Demonstrated
Performance	Response time and system testing
Scalability	Modular architecture and API-based services
Security	Authentication and role-based access
Reliability	Error handling and validation
Usability	User-friendly web/mobile interfaces
Maintainability	Independent modules and modular architecture

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ABILITY OF COMMERCIALIZATION





Start Small. Validate Learning Impact. Scale Across Sri Lankan Education.

THANK YOU…